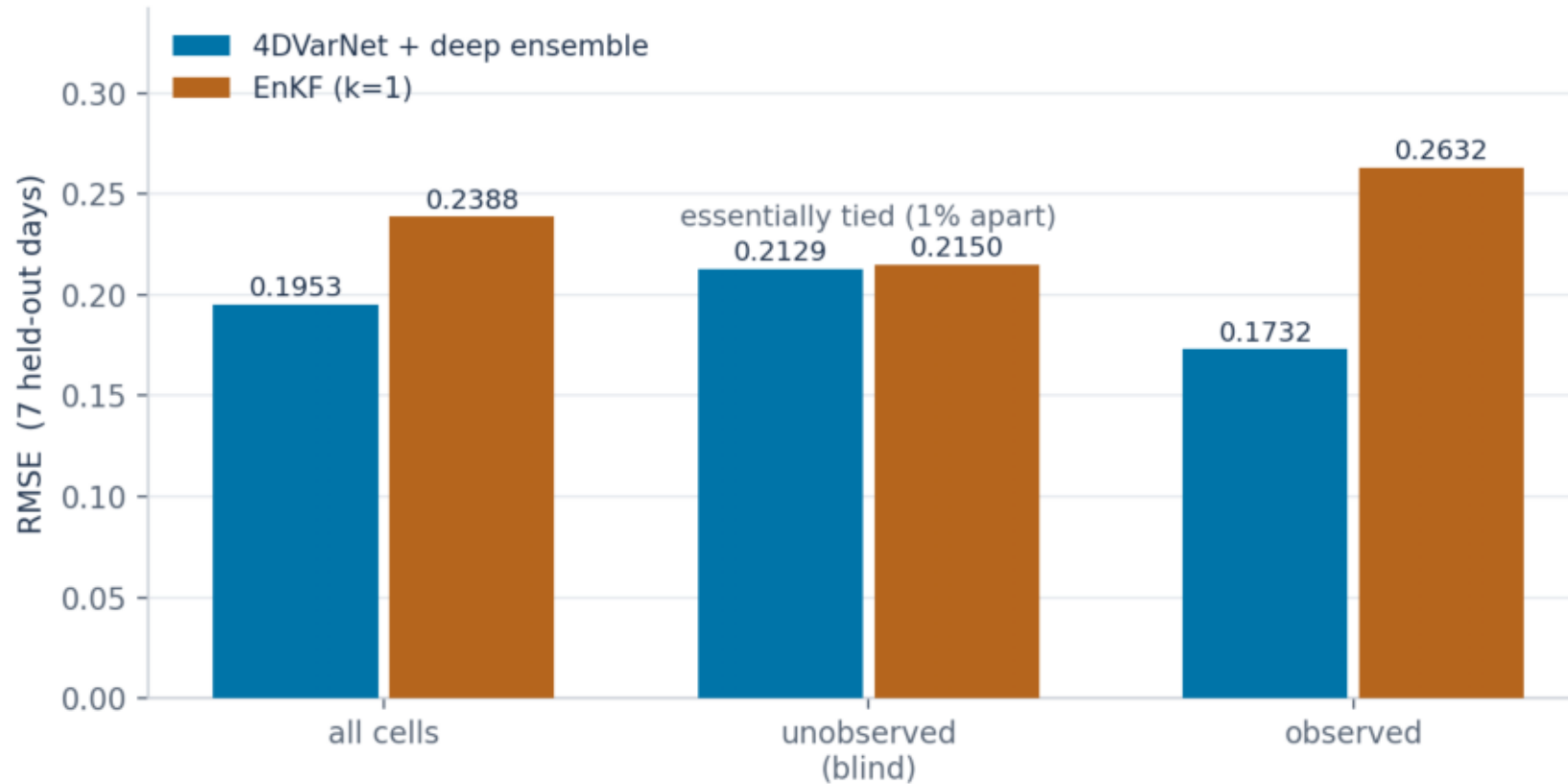


Predictive uncertainty for crowd-field reconstruction

what our σ measures, and what it does not

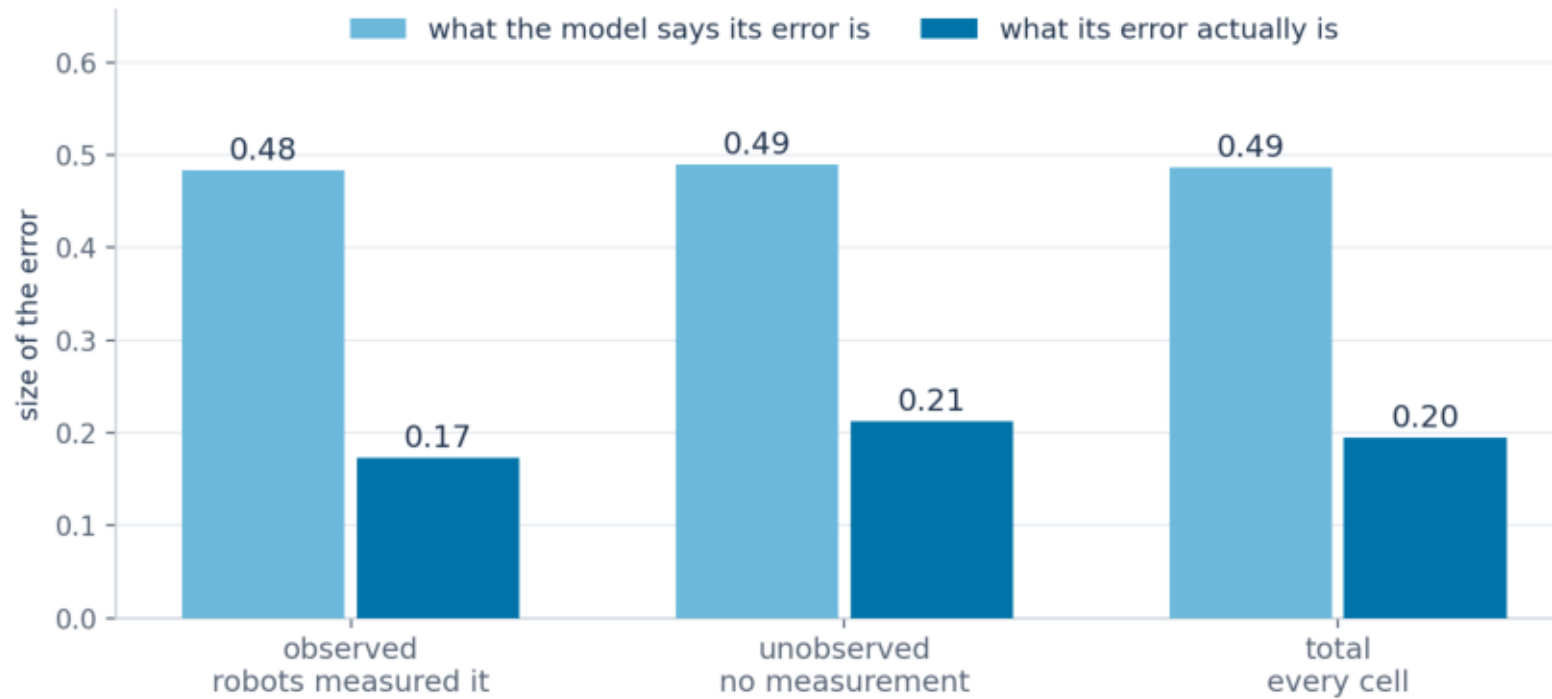
Xinle Zhang

Accuracy — a tie where it counts



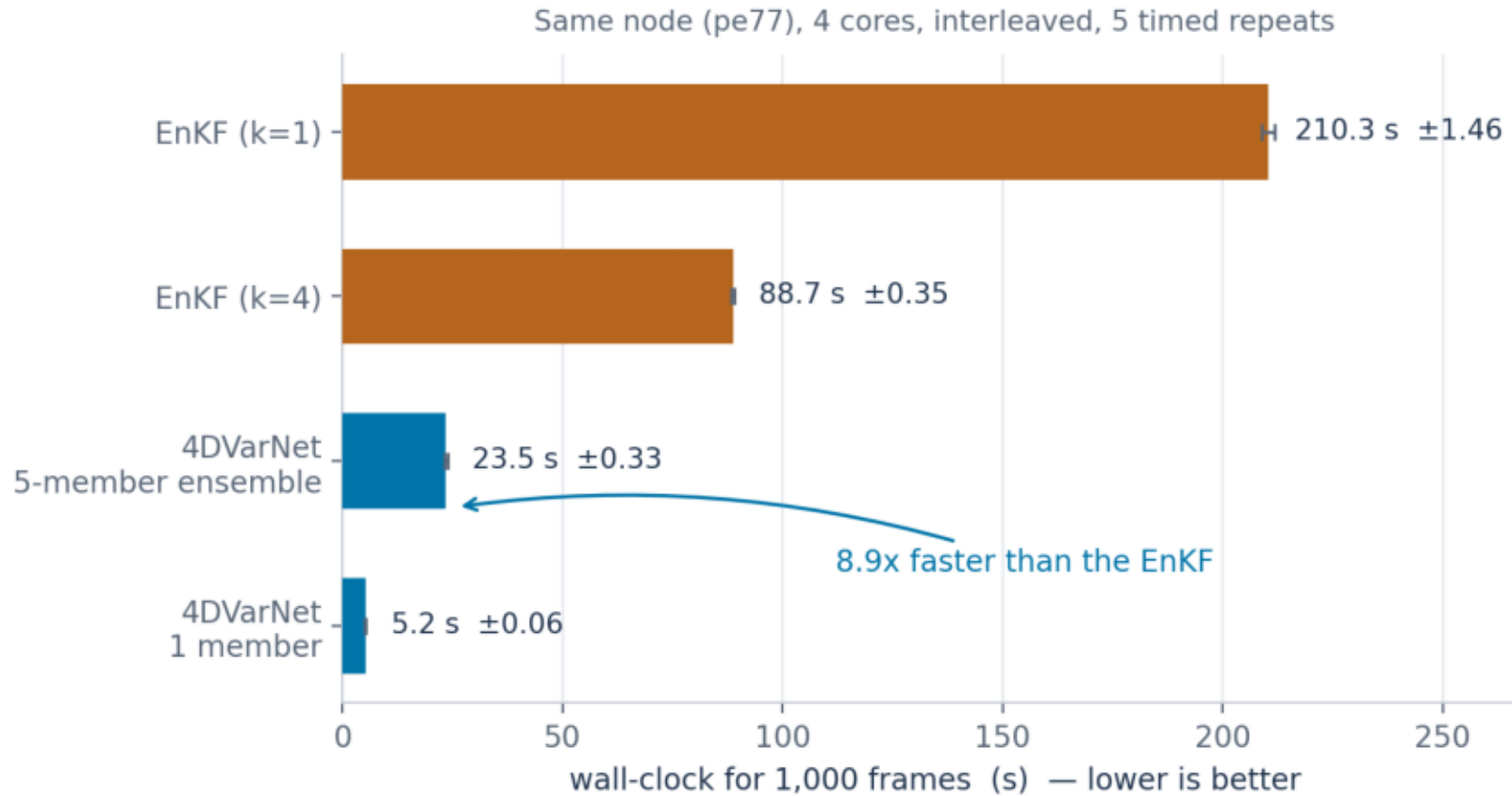
- On unobserved cells the two methods are 1% apart (0.2129 vs 0.2150). That is the region the problem is about, and neither method is ahead there.
- Everything after this slide is about the uncertainty, not the reconstruction.

Is it the right size? It over-states by about 2.5x



- Split by whether a robot actually measured the cell. The model reports almost the same uncertainty either way (0.48 vs 0.49) while the real error differs by 23% (0.17 vs 0.21).
- So it over-states everywhere, and it does not register whether that cell was measured at all.
- Most of the gap comes from the unobserved cells that are truly empty; on the cells that do contain people the two sizes match closely.

Inference time



- 1,000 frames, same node, 4 cores, interleaved order with a warm-up and 5 timed repeats. Largest run-to-run spread 3.6%.
- The 5-network ensemble is 8.9x faster than the EnKF; a single network is 41x faster.